Exam will have 5 problems.

#1 : Computing integrals (substitution rule)

#2: Linear motion (position, velocity, acceleration)

#3: Interpreting a graph

#4: Area

#5: Volume

Any concept from the homework can be on exam.

Bring a calculator (TI-30X IIS only)

Don't bring a note sheet. You will be given a pre-made one during the exam (don't print it out yourself).

Exam is 80 minutes.

A bacterial population starts with 300 bacteria and grows at a rate of $r(t) = (450.207)e^{1.1256t}$ bacteria per hour. How many bacteria will there be after three hours?

Let $p(t)$ = population at time $t$. By net change theorem,

$$p(3) - p(0) = \int_0^3 r(t)\,dt = \int_0^3 ae^{bt}\,dt$$

Method 1: u-substitution

$u = bt$

$du = b\,dt$

$$\int_{t=0}^{t=3} a e^{bt}\,dt = \int_{u=0}^{u=3b} a e^{u} \cdot \frac{1}{b}\,du$$

$$= \frac{a}{b}\int_{0}^{3b} e^{u}\,du$$

$$= \frac{a}{b} e^{u}\Big|_{0}^{3b}$$

$$= \frac{a}{b}\left(e^{3b} - 1\right)$$

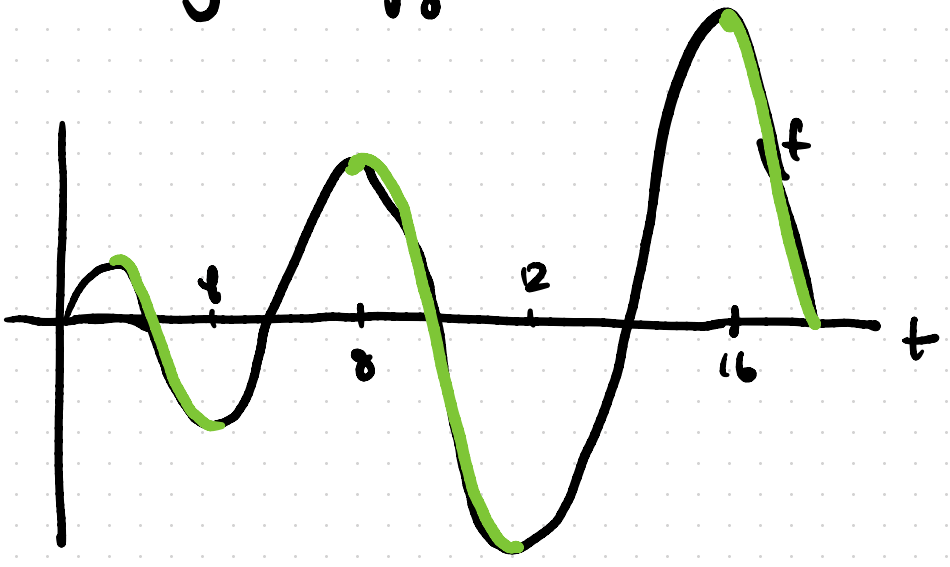
$$\boxed{p(3) = p(0) + \frac{a}{b}\left(e^{3b} - 1\right)}$$

Method: Eyeball it

$$\int_0^3 a e^{bt}\,dt = \frac{a}{b} e^{bt}\Big|_0^3$$

$$= \frac{a}{b}\left(e^{3b} - 1\right)$$

Let $g(x) = \int_0^x f(t)\,dt$, where $f$ is




where are the local maxima of $g$?

2, 10

Where are the local minima of $g$?

6, 14

Where does $g$ attain its absolute maximum?

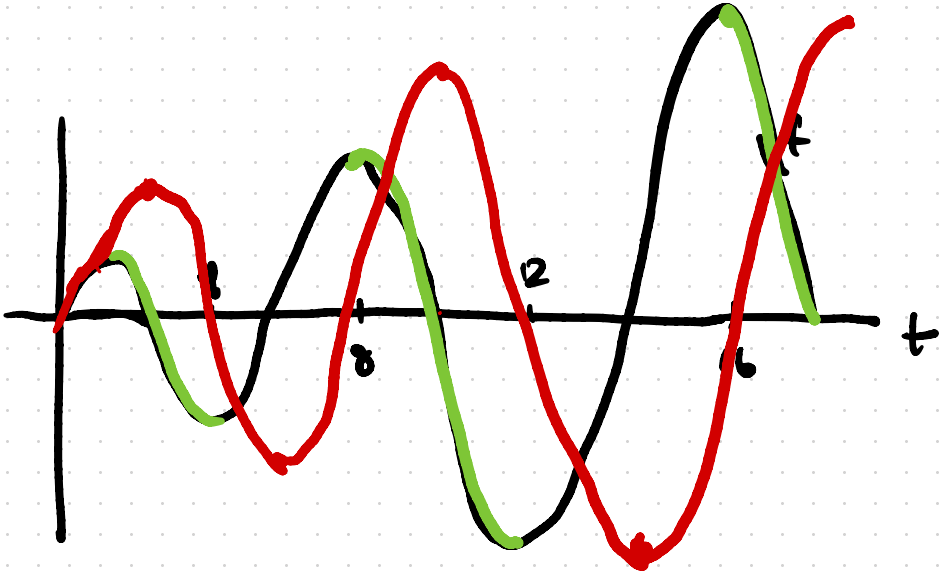
0, 2, 10, 18 18

Where does $g$ attain its absolute minimum? 14

On what intervals is $g$ concave downward?

$g'' < 0$ $\quad g' = f$

$f' < 0$, i.e. $f$ is decreasing



Let $f(x) = \sin x^2$. Write

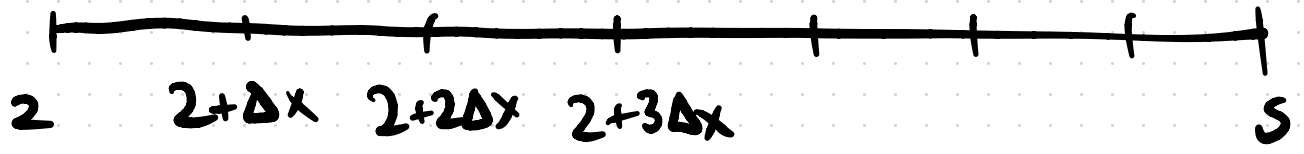
$$\int_2^5 f(t)\,dt$$

as the limit of a sum.

$$\int_2^5 f(t)\,dt = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

where we subdivide $[2, 5]$ into $n$ intervals of length $\Delta x$, and $x_i^*$ is any value chosen in the $i^{th}$ interval.

$$\Delta x = \frac{5-2}{n} = \frac{3}{n}$$




1st interval : $[2, 2+\Delta x]$

2nd : $[2+\Delta x, 2+2\Delta x]$

3rd : $[2+2\Delta x, 2+3\Delta x]$

$\vdots$

$x_i^* = 2 + i\Delta x$

$= 2 + i\left(\frac{3}{n}\right)$

If we choose $x_i^*$ to be the right endpoint of the $i$th interval

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(2+\frac{3i}{n}\right)\frac{3}{n}$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \sin\left(2+\frac{3i}{n}\right)^2 \frac{3}{n}$$